

CDAC FUTURE SKILLS PROGRAMME

Demystifying Android Application Security

Day 02 – Assignment: Android Dynamic Analysis

ADB and Frida – Standard Practical Tasks

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Part A – ADB: Application Setup, Information & Logs

Task 1 – Emulator Detection via ADB

Command:

```
adb devices
```

Output:

```
PS C:\Users\KUDSIT> adb devices
List of devices attached
emulator-5554    device
```

Observation: The configured Pixel Android emulator was successfully detected by ADB with the status `device`, confirming that the workstation can communicate with the emulator.

Screenshot

ADB successfully detecting and communicating with the Pixel emulator.



```
PS C:\Users\KUDSIT> adb devices
List of devices attached
emulator-5554    device
```

Task 2 – Basic Device Properties

Commands:

```
adb shell getprop ro.product.model
adb shell getprop ro.build.version.release
adb shell getprop ro.build.version.sdk
adb shell getprop ro.product.cpu.abi
```

Property	Result
Device model	sdk_gphone64_x86_64
Android version	13
API level	33
CPU architecture	x86_64

Observation: The emulator was identified as `sdk_gphone64_x86_64`, running Android 13 with API level 33 and an x86_64 CPU architecture.

Screenshot

Android emulator device model, Android version, API level, and CPU architecture retrieved using ADB.

```
PS C:\Users\KUDSIT> adb shell getprop ro.product.model
sdk_gphone64_x86_64
PS C:\Users\KUDSIT> adb shell getprop ro.build.version.release
13
PS C:\Users\KUDSIT> adb shell getprop ro.build.version.sdk
33
PS C:\Users\KUDSIT> adb shell getprop ro.product.cpu.abi
x86_64
PS C:\Users\KUDSIT> |
```

Task 3 – Install and Identify the APK

Command 1 – Install the APK:

```
adb install ".\InsecureShop.apk"
```

Output: Performing Streamed Install / Success

Command 2 – Identify the package name:

```
adb shell pm list packages | Select-String "insecure"
```

Output: package:com.insecureshop

Observation: The instructor-provided InsecureShop.apk was successfully installed on the Android emulator. The package manager was then queried to identify the application's full package name, which was found to be `com.insecureshop`.

Item	Result
APK	InsecureShop.apk
Installation	Successful
Package name	com.insecureshop

Screenshot 1

InsecureShop.apk successfully installed on the Android emulator using ADB.

```
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb install ".\InsecureShop.apk"
Performing Streamed Install
Success
```

Screenshot 2

ADB package manager output identifying the InsecureShop package as com.insecureshop.

```
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell pm list packages | Select-String
"insecure"

package:com.insecureshop
```

Task 4 – Package Details and Installed Application Information

The installed package was inspected using Android Package Manager and dumpsys package. The APK was located under the application's `/data/app/` directory. The application version was identified as 1.0 with version code 1, targeting SDK 29. The package information contained the **DEBUGGABLE** flag, indicating that the installed application is debuggable. The application requested several permissions, including `android.permission.INTERNET`, which was granted. Registered components included `ProductListActivity` and `InsecureShopProvider`.

Item	Result
Installed APK path	<code>/data/app/~~Tl5VnP4T4yOrCx1lLZY5Iw==/com.insecureshop-GsoldMwLHXAX2y_BTMoFJQ==/base.apk</code>
Version	1.0
Version code	1
Debuggable	Yes (DEBUGGABLE)
Target SDK	29
Permission	<code>android.permission.INTERNET</code> — granted
Component	<code>com.insecureshop/.ProductListActivity</code>
Additional component	<code>com.insecureshop/.contentProvider.InsecureShopProvider</code>

Commands used

APK path:

```
adb shell pm path com.insecureshop
```

Version / debugging information:

```
adb shell dumpsys package com.insecureshop | Select-String
"versionName|versionCode|debuggable"
```

Permissions:

```
adb shell dumpsys package com.insecureshop | Select-String "permission"
```

Components:

```
adb shell dumpsys package com.insecureshop | Select-String
"Activity|Service|Receiver|Provider"
```

Debuggable confirmation:

```
adb shell dumpsys package com.insecureshop | Select-String "DEBUGGABLE"
```

Screenshot 1

Installed APK path retrieved using ADB (pm path).

```
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell pm path com.insecureshop
package:/data/app/~~Tl5VnP4T4yOrCx1lLZY5Iw==/com.insecureshop-Gs0ldMwLHXAX2y_BTMoFJQ=
=/base.apk
```

Screenshot 2

Version, version code, and debuggable flag retrieved using dumpsys package.

```
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell dumpsys package com.insecureshop
| Select-String "versionName|versionCode|debuggable"

versionCode=1 minSdk=16 targetSdk=29
versionName=1.0
flags=[ DEBUGGABLE HAS_CODE ALLOW_CLEAR_USER_DATA ALLOW_BACKUP ]
pkgFlags=[ DEBUGGABLE HAS_CODE ALLOW_CLEAR_USER_DATA ALLOW_BACKUP ]
```

Screenshot 3

Installed application permissions retrieved using ADB, including android.permission.INTERNET (granted).

```
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell dumpsys package com.insecureshop | Select-String "permission"

Permissions:
Permission [com.insecureshop.permission.READ] (1ce0153):
perm=PermissionInfo{21dbcd4 com.insecureshop.permission.READ}
installPermissionsFixed=false
declared permissions:
  com.insecureshop.permission.READ: prot=normal, INSTALLED
requested permissions:
  android.permission.INTERNET
  android.permission.READ_EXTERNAL_STORAGE
  android.permission.WRITE_EXTERNAL_STORAGE
  android.permission.READ_CONTACTS
  android.permission.WAKE_LOCK
  android.permission.POST_NOTIFICATIONS
  android.permission.READ_MEDIA_AUDIO
  android.permission.READ_MEDIA_VIDEO
  android.permission.READ_MEDIA_IMAGES
install permissions:
  android.permission.INTERNET: granted=true
  android.permission.WAKE_LOCK: granted=true
runtime permissions:
  android.permission.POST_NOTIFICATIONS: granted=false, flags=[
USER_SENSITIVE_WHEN_GRANTED|USER_SENSITIVE_WHEN_DENIED]
  android.permission.READ_EXTERNAL_STORAGE: granted=false, flags=[
USER_SENSITIVE_WHEN_GRANTED|USER_SENSITIVE_WHEN_DENIED|RESTRICTION_INSTALLER_EXEMPT]
  android.permission.READ_MEDIA_IMAGES: granted=false, flags=[
USER_SENSITIVE_WHEN_GRANTED|USER_SENSITIVE_WHEN_DENIED]
  android.permission.READ_MEDIA_AUDIO: granted=false, flags=[
USER_SENSITIVE_WHEN_GRANTED|USER_SENSITIVE_WHEN_DENIED]
  android.permission.READ_MEDIA_VIDEO: granted=false, flags=[
USER_SENSITIVE_WHEN_GRANTED|USER_SENSITIVE_WHEN_DENIED]
  android.permission.WRITE_EXTERNAL_STORAGE: granted=false, flags=[
USER_SENSITIVE_WHEN_GRANTED|USER_SENSITIVE_WHEN_DENIED|RESTRICTION_INSTALLER_EXEMPT]
  android.permission.READ_CONTACTS: granted=false, flags=[
USER_SENSITIVE_WHEN_GRANTED|USER_SENSITIVE_WHEN_DENIED]
```

Screenshot 4

Android components registered by the InsecureShop application (activities and content providers).

```
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell dumpsys package com.insecureshop
| Select-String "Activity|Service|Receiver|Provider"
```

Activity Resolver Table:

```
e93c548 com.insecureshop/.ChooserActivity filter ab55106
e93c548 com.insecureshop/.ChooserActivity filter ab55106
e93c548 com.insecureshop/.ChooserActivity filter ab55106
e93c548 com.insecureshop/.ChooserActivity filter ab55106
e93c548 com.insecureshop/.ChooserActivity filter ab55106
e93c548 com.insecureshop/.ChooserActivity filter ab55106
e93c548 com.insecureshop/.ChooserActivity filter ab55106
e93c548 com.insecureshop/.ChooserActivity filter ab55106
e93c548 com.insecureshop/.ChooserActivity filter ab55106
e93c548 com.insecureshop/.ChooserActivity filter ab55106
a4d91d com.insecureshop/.WebViewActivity filter 9c57792
6e9563 com.insecureshop/.WebView2Activity filter e2bf560
e8783c7 com.insecureshop/.ProductListActivity filter cc3edf4
e93c548 com.insecureshop/.ChooserActivity filter 8b71fe1
e93c548 com.insecureshop/.ChooserActivity filter ab55106
```

Registered ContentProviders:

```
com.insecureshop/.contentProvider.InsecureShopProvider:
  Provider{62a2e7d com.insecureshop/.contentProvider.InsecureShopProvider}
com.insecureshop/androidx.core.content.FileProvider:
  Provider{bce1172 com.insecureshop/androidx.core.content.FileProvider}
```

ContentProvider Authorities:

```
[com.insecureshop.file_provider]:
  Provider{bce1172 com.insecureshop/androidx.core.content.FileProvider}
[com.insecureshop.provider]:
  Provider{62a2e7d com.insecureshop/.contentProvider.InsecureShopProvider}
```

Screenshot 5

Confirmation of the *DEBUGGABLE* package flag.

```
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell dumpsys package com.insecureshop
| Select-String "DEBUGGABLE"

flags=[ DEBUGGABLE HAS_CODE ALLOW_CLEAR_USER_DATA ALLOW_BACKUP ]
pkgFlags=[ DEBUGGABLE HAS_CODE ALLOW_CLEAR_USER_DATA ALLOW_BACKUP ]
```


Task 5 – Start, Verify, and Stop Application

Commands:

```
adb shell monkey -p com.insecureshop 1
adb shell pidof com.insecureshop
adb shell am force-stop com.insecureshop
adb shell pidof com.insecureshop
```

Results:

- App launched successfully.
- **PID: 27489**
- App was force-stopped successfully.
- Final `pidof` returned no output – process stopped.

Observation: The InsecureShop application was successfully launched using ADB. Its running process was identified with **PID 27489** and was then stopped using `am force-stop`. A runtime permission dialog appeared during launch.

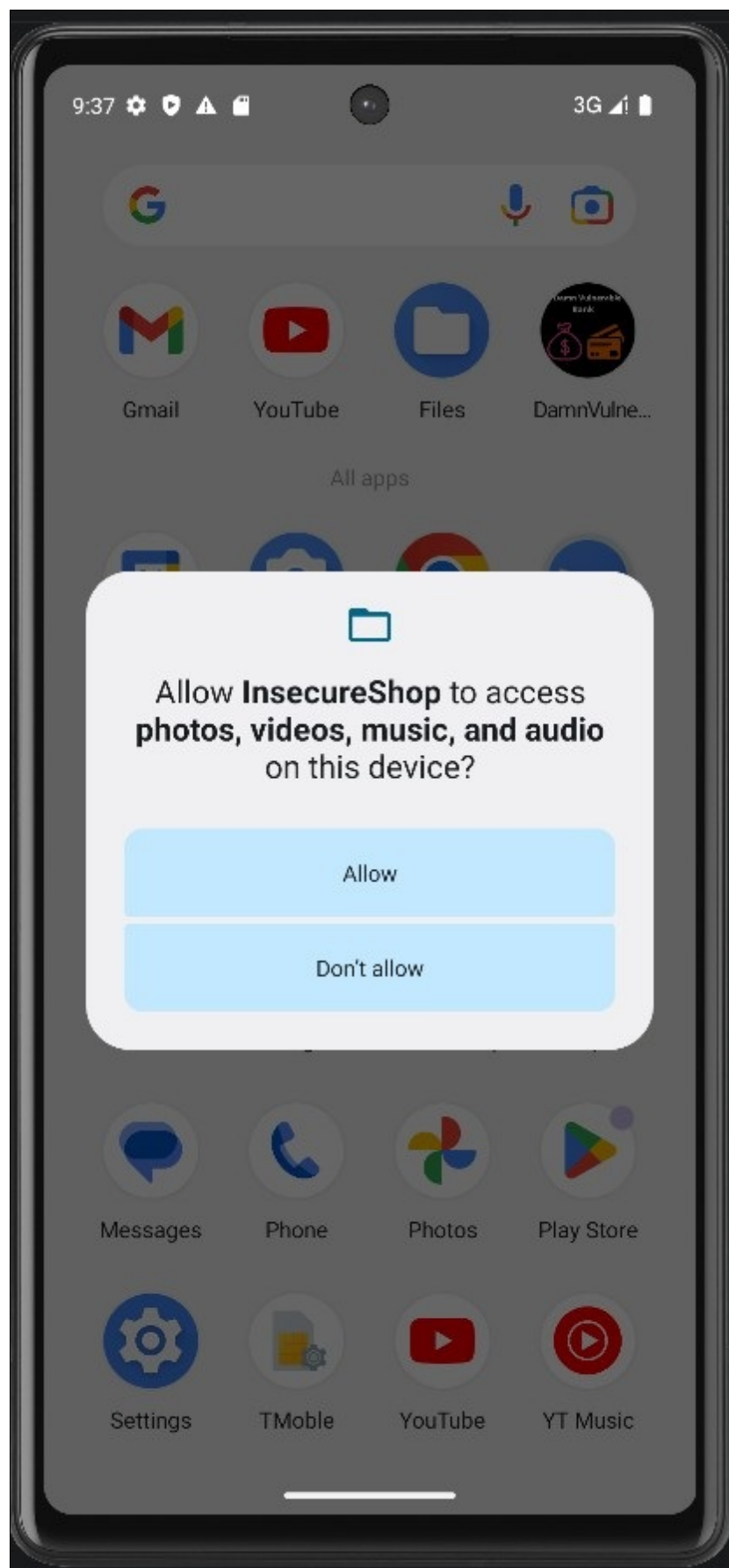
Screenshot 1

Terminal output for launching, checking the PID of, and force-stopping the InsecureShop process.

```
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell monkey -p com.insecureshop 1
  bash arg: -p
  bash arg: com.insecureshop
  bash arg: 1
args: [-p, com.insecureshop, 1]
arg: "-p"
arg: "com.insecureshop"
arg: "1"
data="com.insecureshop"
Events injected: 1
## Network stats: elapsed time=51ms (0ms mobile, 0ms wifi, 51ms not connected)
PS C:\Users\KUDSIT\Downloads\InsecureShop>
PS C:\Users\KUDSIT\Downloads\InsecureShop>
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell pidof com.insecureshop
27489
PS C:\Users\KUDSIT\Downloads\InsecureShop>
PS C:\Users\KUDSIT\Downloads\InsecureShop>
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell am force-stop com.insecureshop
PS C:\Users\KUDSIT\Downloads\InsecureShop>
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell pidof com.insecureshop
```

Screenshot 2

InsecureShop launched through ADB and displaying a runtime permission request.



Task 6 – Private Data Directory Access

Commands:

```
adb shell monkey -p com.insecureshop 1
adb shell run-as com.insecureshop ls
adb shell run-as com.insecureshop pwd
adb shell run-as com.insecureshop ls -la
```

Result: `/data/user/0/com.insecureshop` – accessible directories: `cache`, `code_cache`, `shared_prefs`.

Observation: `run-as` successfully accessed the application's private data directory. The directory `/data/user/0/com.insecureshop` was accessible, and its contents included `cache`, `code_cache`, and `shared_prefs`. This is consistent with the application's **DEBUGGABLE** flag observed in Task 4. (Note: for a non-debuggable package, the expected result would instead be `run-as: package not debuggable` – both outcomes are valid results for this assignment.)

Screenshot

Access to the InsecureShop private data directory using ADB `run-as`, showing `cache`, `code_cache`, and `shared_prefs`.

```
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell monkey -p com.insecureshop 1
  bash arg: -p
  bash arg: com.insecureshop
  bash arg: 1
args: [-p, com.insecureshop, 1]
arg: "-p"
arg: "com.insecureshop"
arg: "1"
data="com.insecureshop"
Events injected: 1
## Network stats: elapsed time=33ms (0ms mobile, 0ms wifi, 33ms not connected)
PS C:\Users\KUDSIT\Downloads\InsecureShop>
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell run-as com.insecureshop ls
cache
code_cache
shared_prefs
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell run-as com.insecureshop pwd
/data/user/0/com.insecureshop
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell run-as com.insecureshop ls -la
total 48
drwx-----  5 u0_a175 u0_a175          4096 2026-09-24 21:33 .
drwxrwx---x 209 system system        12288 2026-09-24 21:23 ..
drwxrws---x  2 u0_a175 u0_a175_cache  4096 2026-09-24 21:23 cache
drwxrws---x  2 u0_a175 u0_a175_cache  4096 2026-09-24 21:23 code_cache
drwxrwx---x  2 u0_a175 u0_a175        4096 2026-09-24 21:33 shared_prefs
PS C:\Users\KUDSIT\Downloads\InsecureShop> |
```

Task 7 – Logcat Capture and Security Keyword Search

Commands:

```
adb logcat -c
$logFile = ".\InsecureShop_logcat.txt"
$logProcess = Start-Process adb -ArgumentList "logcat","-v","time" -RedirectStandardOutput
$logFile -NoNewWindow -PassThru
# ...perform a normal workflow in the app for 30-60 seconds...
Stop-Process -Id $logProcess.Id
Get-Item .\InsecureShop_logcat.txt
Select-String -Path ".\InsecureShop_logcat.txt" -Pattern
"password|token|secret|http://|https://|sqlite"
Select-String -Path ".\InsecureShop_logcat.txt" -Pattern
"userName|username|password|token|secret"
```

Observation 1 – Password logged at runtime

During the normal application workflow, Logcat contained entries tagged `password` that exposed the entered password value. Three entries were observed in the captured log. The actual password value is **redacted** in this report:

```
D/password(27861): [REDACTED]
```

Observation 2 – Password-related runtime activity

Additional Logcat entries showed Android input-method activity associated with password fields in `com.inseureshop`. These entries identified password input fields but did not expose the password value themselves.

Conclusion for Task 7: Logcat was cleared before the workflow and runtime output was captured to `InsecureShop_logcat.txt`. Searching the saved log for security-related keywords produced password-related entries. Most password-related results originated from Android input-method services, while three entries from the application process were tagged `password` and contained the entered value (redacted in this report). This is a strong correlation with the static-analysis finding from `LoginActivity` (`Log.d("password", password);`) – static evidence showed the source code logs the password, and dynamic evidence confirmed the password actually appeared in Logcat at runtime.

Screenshot 1

Logcat search showing password-related runtime entries generated during the InsecureShop workflow, with the password value redacted.

```
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb logcat -c
PS C:\Users\KUDSIT\Downloads\InsecureShop> $logFile = ".\InsecureShop_logcat.txt"
PS C:\Users\KUDSIT\Downloads\InsecureShop> $logProcess = Start-Process adb -ArgumentList "lo
gcat", "-v", "time" -RedirectStandardOutput $logFile -NoNewWindow -PassThru
PS C:\Users\KUDSIT\Downloads\InsecureShop>
PS C:\Users\KUDSIT\Downloads\InsecureShop> Stop-Process -Id $logProcess.Id
PS C:\Users\KUDSIT\Downloads\InsecureShop> Get-Item .\InsecureShop_logcat.txt
```

Directory: C:\Users\KUDSIT\Downloads\InsecureShop

Mode	LastWriteTime	Length	Name
-a----	24-09-2026 21:46	122145	InsecureShop_logcat.txt

```
PS C:\Users\KUDSIT\Downloads\InsecureShop> Select-String -Path ".\InsecureShop_logcat.txt" -
Pattern "password|token|secret|http://|https://|sqlite"
```

```
InsecureShop_logcat.txt:251:09-24 21:45:23.357 W/AutofillSession( 498): failed to
request autofill data for Token{6680fcb ActivityRecord{120c3a8 u0
com.insecureshop/.LoginActivity} t23}}
InsecureShop_logcat.txt:315:09-24 21:45:25.995 D/ContentCapturePerUserService( 498):
Notified activity assist data for activity: Token{6680fcb ActivityRecord{120c3a8 u0
com.insecureshop/.LoginActivity} t23}} without a session Id
InsecureShop_logcat.txt:426:09-24 21:45:37.098 I/GoogleInputMethodService(12050):
GoogleInputMethodService.onStartInput():1877
onStartInput(EditorInfo{inputType=0x81(Password) imeOptions=0x40000005
privateImeOptions=null actionName=NEXT actionLabel=null actionId=0 initialSelStart=0
initialSelEnd=0 initialCapsMode=0x0 hintText=non-empty label=null
packageName=com.insecureshop fieldId=2131296374 fieldName=null extras=null}, false)
InsecureShop_logcat.txt:432:09-24 21:45:37.102 I/GoogleInputMethodService(12050):
GoogleInputMethodService.onStartInputView():1995
onStartInputView(EditorInfo{inputType=0x81(Password) imeOptions=0x40000005
privateImeOptions=null actionName=NEXT actionLabel=null actionId=0 initialSelStart=0
initialSelEnd=0 initialCapsMode=0x0 hintText=non-empty label=null
packageName=com.insecureshop fieldId=2131296374 fieldName=null extras=null}, false)
InsecureShop_logcat.txt:437:09-24 21:45:37.130 I/AndroidIME(12050):
AbstractIme.onActivate():86 PasswordIme.onActivate() : EditorInfo =
inputType=0x81(Password) imeOptions=0x40000005 privateImeOptions=null actionName=NEXT
actionLabel=null actionId=0 initialSelStart=0 initialSelEnd=0 initialCapsMode=0x0
hintText=non-empty label=null packageName=com.insecureshop fieldId=2131296374
fieldName=null extras=null, IncognitoMode = false, DeviceLocked = false
InsecureShop_logcat.txt:457:09-24 21:45:37.166 I/CurrentMicStatusHolder(12050):
CurrentMicStatusHolder.onStartInputView():79 Current Mic status =
{MicIconHidden-SettingOff,MicIconHidden-PasswordOrNumberOrDateInputType,}
InsecureShop_logcat.txt:465:09-24 21:45:39.095 D/InputMethodManager(27861):
```

Screenshot 2

InsecureShop_logcat.txt file created on disk (~120 KB) containing the captured runtime log.

Name	Date modified	Type	Size
▼ Today			
 InsecureShop_logcat	24-09-2026 21:46	Text Document	120 KB

Part B – Frida: Basic Runtime Verification

Task 8 – Frida Client/Server Setup

Commands:

```
frida --version
adb shell getprop ro.product.cpu.abi
adb shell getprop ro.product.cpu.abi.list
where.exe frida
adb shell chmod 755 /data/local/tmp/frida-server
adb shell ls -l /data/local/tmp/frida-server
adb shell "/data/local/tmp/frida-server &"
adb shell ps -A | Select-String "frida-server"
frida-ps -U
```

Requirement	Result
Frida client	17.18.0
Emulator ABI	x86_64
Matching server	17.18.0 Android x86_64
Server pushed	/data/local/tmp/frida-server
Execute permission	755 / -rwxr-xr-x
Server running	PID 7564
frida-ps -U connection	SUCCESS

Observation: Frida client version **17.18.0** was installed on the Windows host. The emulator was identified as **x86_64**, so the corresponding Android x86_64 Frida server was selected. The server was copied to `/data/local/tmp/frida-server`, given execute permission, and started with root privileges. The running server was confirmed using `ps` (PID 7564), and `frida-ps -U` successfully listed processes from the emulator, confirming communication between the Frida client and server.

Screenshot 1

frida-server started (PID 7564) and frida-ps -U listing Android processes on the emulator.


```

PS C:\Users\KUDSIT> adb shell chmod 755 /data/local/tmp/frida-server
PS C:\Users\KUDSIT> adb shell ls -l /data/local/tmp/frida-server
-rwxr-xr-x 1 root root 115966424 2026-09-24 22:04 /data/local/tmp/frida-server
PS C:\Users\KUDSIT> adb shell "/data/local/tmp/frida-server &"

PS C:\Users\KUDSIT> adb shell ps -A | Select-String "frida-server"

root          7564      1 12804680 135480 do_sys_poll          0 S frida-server

PS C:\Users\KUDSIT> frida-ps -U
  PID  Name
-----
2475  Calendar
5689  Camera
3181  Chrome
2827  Clock
2429  Contacts
5531  Drive
5505  Files
6043  Gmail
2119  Google
5828  Maps
5392  Messages
3544  Phone
4475  Photos
1354  SIM Toolkit
1565  Settings
2152  YouTube
2085  YouTube Music
7276  adbd
  175  android.hardware.atrace@1.0-service
  317  android.hardware.audio.service
  318  android.hardware.authsecret@1.0-service
  487  android.hardware.biometrics.face@1.0-service.example
  488  android.hardware.biometrics.fingerprint@2.1-service
  319  android.hardware.bluetooth@1.1-service.btlinux
  323  android.hardware.camera.provider@2.4-service_64
  324  android.hardware.camera.provider@2.7-service-google
  325  android.hardware.cas@1.2-service
  326  android.hardware.contexthub@1.1-service.mock
  327  android.hardware.drm-service.clearkey
  328  android.hardware.drm-service.widevine
  333  android.hardware.drm@1.0-service
  334  android.hardware.drm@1.4-service.clearkey

```

Screenshot 2

Continuation of the frida-ps -U process listing, confirming full communication with the device.

```

6255  com.google.android.permissioncontroller
4015  com.google.android.permissioncontroller

```

```
4017 com.google.android.projection.gearhead:car
5082 com.google.android.projection.gearhead:shared
1973 com.google.android.providers.media.module
6297 com.google.android.settings.intelligence
1377 com.google.android.setupwizard
4997 com.google.android.webview
3152 com.google.android.webview:webview_service
2181 com.google.process.gapps
2025 com.google.process.gservices
 374 credstore
 296 dhcpclient
 413 drmserver
 485 gatekeeper
 378 gpuservice
 451 hvcl
 150 hwservicemanager
 460 incidentd
   1 init
  94 init
 463 installd
 321 ip6tables-restore
 320 iptables-restore
 174 keystore
 479 libgoldfish-rild
 144 lmkd
7568 logcat
 143 logd
 422 mdnssd
 464 media.extractor
 466 media.metrics
 468 mediaserver
 480 mediaswcodec
 311 netd
6873 prng_seeder
7315 process-tracker
 158 qemu-props
 145 servicemanager
```



```
155 sh
310 statsd
470 storaged
382 surfaceflinger
556 system_server
214 tombstoned
441 traced
439 traced_probes
96 ueventd
159 vold
1262 webview_zygote
476 wificond
1069 wpa_supplicant.conf
```

Screenshot 3

Frida client version, emulator CPU ABI/ABI list, and Frida install path on the Windows host.

```
PS C:\Users\KUDSIT\Downloads\InsecureShop> frida --version
17.18.0
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell getprop ro.product.cpu.abi
x86_64
PS C:\Users\KUDSIT\Downloads\InsecureShop> adb shell getprop ro.product.cpu.abi.list
x86_64
PS C:\Users\KUDSIT\Downloads\InsecureShop> where.exe frida
c:\Users\KUDSIT\AppData\Roaming\Python\python313\Scripts\frida.exe
```

Task 9 – Basic Frida Script Test

Pre-checks:

```
adb install "C:\Users\KUDSIT\Downloads\InsecureShop\InsecureShop.apk"
adb shell pm list packages | Select-String "insecure"
frida-ps -U | Select-String "frida-server"
```

Test script (frida_test.js):

```
setTimeout(function () {
    if (Java.available) {
        Java.perform(function () {
            try {
                console.log("[+] Frida attached successfully");
                console.log("[+] Java runtime is available");
            } catch (error) {
                console.log("[-] Error inside Java.perform");
                console.error(error);
            }
        });
    } else {
        console.log("[-] Java is not available in this process");
    }
}, 500);
```

Run command:

```
frida -U -f com.insecureshop -l frida_test.js
```

Key output:

```
Connected to Android Emulator 5554 (id=emulator-5554)
Spawned `com.insecureshop`. Resuming main thread!
[Android Emulator 5554::com.insecureshop ]-> [+] Frida attached successfully
[+] Java runtime is available
```

Observation: Frida successfully connected to the Android emulator and spawned the `com.insecureshop` application. The test script executed inside the application process and confirmed that Frida was successfully attached and that the Android Java runtime was accessible.

Test	Result
Frida attached successfully	Yes
Java runtime available	Yes
Target application	com.insecureshop
Frida version	17.18.0

Screenshot 1

Creation and verification of `frida_test.js` in PowerShell.

```

PS C:\Users\KUDSIT> cd "C:\Users\KUDSIT\Downloads\InsecureShop"
PS C:\Users\KUDSIT\Downloads\InsecureShop> Set-Content -Path .\frida_test.js -Value @(
>> 'setTimeout(function () {'
>> '    if (Java.available) {'
>> '        Java.perform(function () {'
>> '            try {'
>> '                console.log("[+] Frida attached successfully");'
>> '                console.log("[+] Java runtime is available");'
>> '            } catch (error) {'
>> '                console.log("[-] Error inside Java.perform");'
>> '                console.error(error);'
>> '            }'
>> '        });'
>> '    } else {'
>> '        console.log("[-] Java is not available in this process");'
>> '    }'
>> '}, 500);'
>> )
PS C:\Users\KUDSIT\Downloads\InsecureShop> Get-Content .\frida_test.js
setTimeout(function () {
    if (Java.available) {
        Java.perform(function () {
            try {
                console.log("[+] Frida attached successfully");
                console.log("[+] Java runtime is available");
            } catch (error) {
                console.log("[-] Error inside Java.perform");
                console.error(error);
            }
        });
    } else {
        console.log("[-] Java is not available in this process");
    }
}, 500);

```

Screenshot 2

frida -U -f com.insecureshop -l frida_test.js showing successful attachment and Java runtime availability.

```

PS C:\Users\KUDSIT\Downloads\InsecureShop> frida -U -f com.insecureshop -l frida_test.js

  /_ _ |   Frida 17.18.0 - A world-class dynamic instrumentation toolkit
 |C_| |
  >_ _ |   Commands:
 /_/_|_|   help      -> Displays the help system
 . . . .   object?   -> Display information about 'object'
 . . . .   exit/quit -> Exit
 . . . .
 . . . .   Prefer a GUI? Luma is the official Frida app, with a live REPL,
 . . . .   persistent sessions & collaboration. https://luma.frida.re/
 . . . .
 . . . .   Connected to Android Emulator 5554 (id=emulator-5554)
Spawned `com.insecureshop`. Resuming main thread!
[Android Emulator 5554::com.insecureshop ]-> [+] Frida attached successfully
[+] Java runtime is available

```

Conclusion

This practical provided hands-on experience with ADB-based Android application analysis and Frida-based runtime instrumentation. The InsecureShop APK was successfully installed, its package information and runtime behaviour were examined, and Logcat was used to identify security-relevant application output. One issue encountered was that the original Android emulator used a production build where `adb root` was unavailable, which prevented Frida Server from starting correctly. A new Android 13 Google APIs emulator was therefore configured, allowing root access and successful Frida Server execution. Another issue was initially encountered while extracting the `.xz` Frida Server archive, which was resolved using 7-Zip. Finally, Frida 17.18.0 successfully connected to the emulator, spawned InsecureShop, and confirmed Java runtime availability. Overall, the practical demonstrated the difference between static inspection and runtime analysis and provided experience with ADB, Logcat, Frida Server, and basic Frida instrumentation.